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Specialization: *Healthcare*

Business Focus: *Public Health Analysis*

Tool: *Power BI*

GLOBAL HIV TRENDS AND TREATMENT EFFECTIVENESS ANALYSIS

Project Learning Opportunities

This project provides hands-on experience in:

- *Time-Series Analysis : Understanding disease trends over time*
- *Data Modeling (Star Schema) : Structuring healthcare data for analysis*
- *KPI Development : Creating meaningful health indicators (death rate, growth rate)*
- *Correlation Analysis : Evaluating relationship between treatment (ART) and outcomes*
- *Data Storytelling : Translating complex health data into actionable insights*

Tools and Technology to be Used

Power BI

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Project Overview

Introduction to the Project

Over the past three decades, WHO have made significant investments in combating the spread and impact of HIV. Through a combination of prevention programs, awareness campaigns, and the introduction of Antiretroviral Therapy (ART), HIV has transitioned from a fatal disease to a manageable chronic condition for many individuals.

Despite these efforts, HIV remains a major public health concern, particularly in regions with limited healthcare infrastructure. While global reports suggest declining mortality rates, questions remain about the true effectiveness and sustainability of current interventions.

In this project, the data team is tasked with evaluating:

- Whether HIV infections are truly declining
- If increased treatment access is reducing mortality
- How the overall burden of HIV is evolving globally

This analysis should provides a holistic, data-driven view of the global HIV landscape.

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Project Overview

Project Background

WHO face challenges such as:

- Uneven distribution of healthcare resources
- Inconsistent data reporting across countries
- Difficulty measuring long-term intervention impact

The introduction of ART has significantly improved survival rates, but:

- New infections still occur
- Regional disparities persist
- Healthcare systems face increasing long-term burden

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Project Overview

Problem Statement

Despite the availability of HIV-related data, stakeholders lack a unified analytical framework to:

1. Evaluate the effectiveness of treatment programs
2. Understand the relationship between infections, deaths, and treatment
3. Identify regions requiring urgent intervention

This limits strategic planning and resource allocation.

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Rationale for the Project

1.

Disease Monitoring

Track global HIV spread over time

2.

Evaluate treatment impact

Access how ART coverage affects mortality

3.

Support strategic decisions

Enable policymakers to target high risk regions

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Data Description

Dataset Includes:

1. Country: Represents the geographic location where the data was recorded

2. Year: Represents the time dimension (annual data)

3. Indicator: Defines the type of HIV metric being recorded:

- New Infections: Number of newly reported HIV cases...Measures disease spread (incidence)
- Deaths: Number of deaths caused by HIV...Measures severity and mortality impact
- People Living with HIV: Total number of individuals currently living with HIV...Measures disease burden (prevalence)
- ART Coverage (%): Percentage of HIV-positive individuals receiving treatment...Measures treatment access and healthcare effectiveness

4. Value: Numeric value corresponding to each indicator

Interpretation depends on the indicator:

- Counts (infections, deaths, population)
- Percentage (ART coverage)

Data Characteristics:

- Multi-country
- Time-series
- Aggregated annually

Business Analysis Questions

- How many new HIV infections are occurring over time?
- What is the total number of HIV-related deaths globally?
- How many people are currently living with HIV?
- What is the overall level of treatment coverage across countries?
- What is the mortality and survival rate of the population
- For every 1,000 new HIV infections, which countries record the highest number of deaths?
- How does the trend of infections compare with deaths?
- Which regions are most affected by HIV?
- Which countries have the best treatment access?
- Do high-infection countries have adequate treatment coverage?
- Does higher ART coverage correlate with lower mortality?
- Which countries are most at risk due to poor treatment access?

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Tech Stack

Power BI

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Project Workflow

DATA COLLECTION

Retrieve the data from Google sheets

DATA MODELLING

Transform and model the dataset into Facts and dimension tables. 1 facts and 3 Dimension tables

DATA ABALYSIS AND VISUALIZATION

Analyze the dataset and visualize your data to show actionable insights

RECOMMENDATIONS

Based on the analysis, provide actionable recommendations to help policy makers and government make informed decisions.

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Data source and Dataset

Data source : WHO Official Website

Right click the link and copy the Google Sheet link below to access the dataset 👇 👇 👇

https://docs.google.com/spreadsheets/d/1G-9L46KUTepJLf2SFL_Ah6XeZDgWzoaEZ--CHQfNXJk/edit?usp=sharing

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Project Submission Direction

Dashboard Requirements

Your Dataset must contain a cleaned Star Schema Data Model showing your fact table as well as your dimension tables.

Dashboard Requirements:

- Create your clean and intuitive dashboard answering the key business questions
- Include important KPIs like death rate, survival rate, YOY growth(Optional but impressive) etc.
- Your dashboard must include clean and professional layout, with interactive slicers on year, location or indicators etc

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Project Submission direction

Submission Directions

After creating your dashboard, take screenshot of your key charts or graphs and attach them to your presentation slides.

1. Use Data Storytelling to explain your insights on the present condition of global HIV status in the world. Make sure to attach relevant chart or graphs that will aid your communication.
2. Give your actionable recommendation at the end of your presentation
3. Convert your PowerPoint slide into a PDF format
4. Submit ONLY the PDF format on your Google classroom.

Don’t forget to post your work on LinkedIn and tag 10Alytics.
If you have challenges please reach out to your POD Members or Class Coordinator.
Wishing you all the Best!!!